Noah Stein

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Education

University of Michigan, *College of Engineering* **Ann Arbor, MI**

B.S.E. in Computer Engineering, Minor in Mathematics, *Aug. 2022 - May 2026*

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Assembly, ML/AI, Databases, Computer Architecture, Digital Integrated Circuits, Signals and Systems, Verilog/RTL Design, Feedback Control, Electronics, FPGA/HDL Courses, Linear Algebra, Statistics and Data Analysis, Artificial Intelligence, Machine Learning.

Project Experience

Brain Wave Controlled Light Switch **Ann Arbor, MI**

Position - Technical Lead & Lead BCI Developer *May 2025 (completed)*

- Built a real-time brain–computer interface that streamed EEG data from a NeuroSky MindWave headset into a Python control pipeline, enabling fully wireless, real-time actuation of a Wi-Fi–connected light switch.
- Executed custom EEG packet parsing, signal smoothing, and adaptive thresholding algorithms, integrated with ESP32 firmware over serial communication to trigger GPIO-controlled relay actuation; calibrated user-specific thresholds across 20+ trials, improving activation consistency by ~25% while minimizing noise-induced false positives.

Brain Wave Machine Learning Simulation **Ann Arbor, MI**

Position - Lead Machine Learning Engineer & BCI Research Developer *May 2025 (completed)*

- Developed a deep learning pipeline to classify multiple limb motor imagery EEG signals with ~80–85% accuracy.
- Implemented an EEGNet-style CNN in Python PyTorch and optimized hyperparameters such as learning rate, dropout, and filter size. Achieved a ~15% accuracy improvement over commonly used baseline models.
- Built a real-time inference pipeline achieving <200 ms end-to-end latency and integrated prediction smoothing with confidence gating, enabling stable, smooth multi-joint prosthetic simulation while reducing jitter and unstable predictions by ~30%.

Advanced Microarchitecture Processor Project

- Designed and accomplished a 2-wide superscalar out-of-order processor with register renaming and speculative execution.
- Implemented 10+ RTL modules including a physical register file, free list, rename tables, reorder buffer, micro-op queue, branch predictor/BTB, and pipeline issue logic; added an instruction cache and optimized control paths, reducing structural stalls by ~20% in benchmark tests.
- Collaborated within a 5-person processor design team, owning RTL modules and directing cross-stage integration, debug, and Verilog testbench verification to resolve renaming, forwarding, and speculation recovery edge cases.

Digital Integrated Circuits Designs (Virtuoso PMOS Schematics)

- Designed and simulated CMOS circuits, such as an 8-bit adder (inverters, NAND/NOR, PTL, keeper circuits, latches, domino logic) and evaluated switching thresholds, propagation delay, and noise margins.
- Performed transient and parametric analyses and honed transistor sizing, improving propagation delay by ~15–20%, and simulated analog envelope detectors and IF filters using Spectre SPICE, analyzing charge leakage and metastability behavior.

Work Experience

University of Michigan Athletics **Ann Arbor, MI**

Facility Assistant *May 2024 - Present*

- Monitored and provided operational assistance during live athletic events by managing logistics, setup, and breakdown activities, maintaining high attention to detail and reliability in fast-paced, time-critical environments.
- Collaborated closely with supervisors, coworkers, coaches, and event staff to execute events smoothly, requiring frequent, clear communication and rapid adaptation to changing operational conditions and training for new employees

Monroe Golf and Country Club **Monroe, MI**

Course Management Assistant *May 2020 - Aug. 2024*

- Managed course equipment and customer support; strengthened communication, organizational, and operational skills by leveraging effective communication with superiors and 60+ members while coordinating daily operations and events.

Skills & Interests

Programming Languages: C/C++, Assembly, Bash, Python, MATLAB, HTML/CSS/JavaScript, Excel, PowerPoint.

Hardware/EE: SystemVerilog, FPGA Design, Digital Logic Design, Cadence Virtuoso, SPICE, Embedded Systems, Microcontrollers.

Software/Tools: Git, GitHub, Linux, VS Code, PyTorch/Tensorflow, LTSpice, ModelSim.

Concepts: Computer Architecture, Data Structures & Algorithms, Machine Learning, Signals and Systems, Control Systems, Quantum Computing Fundamentals, Dynamic Programming.

Interpersonal: Project Management, Leadership, Communication, Collaboration, Organization, Decision-making, Problem Solving, Strategic Planning.

Extracurricular: Theta Xi Fraternity Sigma Chapter, The Michigan Daily, Piano, Guitar, Intramural Basketball and Flag Football.